

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Industry Problem Task (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO3	Assessment:	Assessment 2: Industry Problem Task
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I T Bhupal/192425243 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: bhupalreddy

QUESTIONS: 5

Total Marks: 50

Annexure A

Industry Problem Task

1. Industrial Defect Detection

A manufacturing company wants to automatically identify defective products from conveyor belt images. Design a transfer learning-based defect detection/classification solution using DenseNet or another suitable pre-trained CNN. Explain your model selection, feature extraction, fine-tuning strategy, and evaluation metrics.

2. Medical Diagnosis Support System

A hospital wants an AI system to assist doctors in identifying diseases from medical images. Develop a deep learning solution using transfer learning for medical image classification. Compare feature extraction and fine-tuning approaches and justify the final model.

3. Customer Review Sentiment Analysis

An e-commerce company receives thousands of customer reviews every day and wants to automatically understand customer sentiment. Design an RNN/LSTM/BiRNN-based sentiment analysis system to classify reviews as positive, negative, or neutral.

4. Predictive Text for a Mobile Application

A mobile keyboard company wants to provide intelligent next-word suggestions. Develop an LSTM-based next-word prediction model and explain how sequential data is processed using BPTT and how LSTM handles long-term dependencies.

5. Automated Customer Support

A multinational company wants to automatically translate customer queries between languages. Design an Encoder–Decoder-based Sequence-to-Sequence model for machine translation. Explain the role of the encoder, decoder, and sequence representation.

6. Social Media Content Monitoring

A social media platform wants to automatically classify posts based on sentiment or category. Develop and compare RNN, BiRNN, and LSTM models for the problem. Analyze which architecture is most suitable for understanding contextual information.

7. Agriculture Crop Disease Detection

An agricultural technology company wants to identify crop diseases from leaf images submitted by farmers. Develop a transfer learning-based image classification system using DenseNet. Apply data augmentation, feature extraction, and fine-tuning to improve performance.

8. Financial News Analysis

A financial organization wants to analyze news headlines and predict whether the market sentiment is positive, negative, or neutral. Design an LSTM or BiRNN-based sequential learning model and compare its performance with a basic RNN.

9. Automated Document Processing

A company receives a large number of scanned documents and wants to automatically classify them into categories such as invoices, resumes, reports, and contracts. Design a deep learning-based document classification system using transfer learning. Discuss preprocessing, feature extraction, model development, and fine-tuning.

10. Smart Traffic Monitoring

A smart city project wants to classify images captured from traffic cameras into categories such as cars, buses, motorcycles, and emergency vehicles. Develop a transfer learning-based traffic classification model using a pre-trained DenseNet architecture and optimize it for real-time deployment.

1. Industrial Defect Detection

Design: Transfer Learning-based classification using DenseNet.

Model Selection:

DenseNet (Densely Connected Convolutional Network) is an ideal choice for defect detection. Unlike standard CNNs, DenseNet connects each layer to every other layer in a feed-forward fashion. This dense connectivity encourages heavy feature reuse, reduces the number of parameters, and mitigates the vanishing gradient problem. This is critical for industrial imaging, where defects (like microscopic cracks or scratches) require the model to capture fine, granular textures without losing information as it passes through deep network layers.

Feature Extraction:

In this phase, we leverage the pre-trained weights (typically trained on large datasets like ImageNet). We remove the original classification head and freeze the convolutional base. The base acts as a powerful feature extractor, identifying basic edges, shapes, and textures from the conveyor belt images.

Fine-Tuning Strategy:

Because industrial materials look vastly different from standard ImageNet pictures (dogs, cats, cars), pure feature extraction is not enough. The strategy is to:

1. Train a new custom classification head on top of the frozen base.
2. Unfreeze the last dense block of the pre-trained DenseNet.
3. Retrain these top layers alongside the classification head using a very low learning rate. This adapts the higher-level, abstract features to the specific visual domain of the industrial products.

Evaluation Metrics:

Because defective products are usually rare compared to normal products, there is a heavy class imbalance. Accuracy is a misleading metric here. The model must be evaluated using **Precision**, **Recall** (Sensitivity), **F1-Score**, and the **AUC-ROC curve**. High **Recall** is prioritized to ensure no defective product slips through the system (minimizing false negatives).

2. Medical Diagnosis Support System

Design: Deep Learning Image Classification using Transfer Learning.

Feature Extraction vs. Fine-Tuning:

- **Feature Extraction:** Involves locking the pre-trained convolutional layers so their weights cannot update during training. Only a newly added, randomly initialized output layer is trained. This is computationally cheap and prevents overfitting on small medical datasets, but it struggles because medical scans (X-rays, MRIs) lack the visual similarity to everyday images used in pre-training.
- **Fine-Tuning:** Involves unfreezing the later layers (or the entire network) of the pre-trained model and updating their weights during training. This allows the network to learn domain-specific features, such as the subtle contrast differences of a tumor or a bone fracture.

Final Model Justification:

The final model should utilize a **Fine-Tuning** approach. Medical imaging requires highly specialized pattern recognition. While the early layers of a pre-trained model can safely remain frozen (as they detect universal features like lines and basic edges), the later convolutional blocks must be unfrozen and fine-tuned. This strikes the best balance: it leverages pre-existing knowledge to save training time while adapting the model's deeper layers to accurately identify complex, disease-specific biological markers.

3. Customer Review Sentiment Analysis

Design: Sentiment Analysis System using a Bidirectional LSTM (BiLSTM).

System Architecture:

To classify reviews into positive, negative, or neutral categories, a BiLSTM is the most effective recurrent architecture. The pipeline consists of:

1. **Text Preprocessing & Tokenization:** Cleaning the reviews and converting words into integer tokens.
2. **Word Embedding Layer:** Mapping tokens into dense, continuous vectors (e.g., Word2Vec or GloVe) to capture semantic meanings.
3. **BiLSTM Layer:** Processing the sequence of embeddings.
4. **Dense/Classification Layer:** A fully connected layer with a Softmax activation function to output probabilities for the three classes (Positive, Negative, Neutral).

Why BiLSTM:

Standard RNNs or LSTMs read text strictly from left to right. However, sentiment often depends on context found later in a sentence (e.g., "The food was not exactly what I would call *good*"). A Bidirectional LSTM utilizes two hidden layers to process the text in both forward and backward directions simultaneously. By concatenating the outputs of both passes, the model captures the full context of a word based on its entire surrounding environment, leading to highly accurate sentiment classification.

4. Predictive Text for a Mobile Application

Design: Next-Word Prediction Model using LSTM.

Processing Sequential Data with BPTT:

To train the model on sequential text, it relies on Backpropagation Through Time (BPTT). When a sentence is fed into the network, the model unrolls over a series of time steps (one step per word). During training, the model makes a prediction at each step, and a loss is calculated. BPTT works by propagating this error backwards through the unrolled time steps, updating the network weights to minimize the error of future predictions.

Handling Long-Term Dependencies:

Standard Recurrent Neural Networks (RNNs) struggle with long sequences because gradients shrink exponentially during BPTT (the "vanishing gradient" problem), causing the model to forget earlier words in a sentence. LSTMs (Long Short-Term Memory networks) solve this using a central "Cell State"—a memory highway that runs straight down the entire chain, allowing gradients to flow unchanged.

The LSTM regulates this memory using three mathematical gates:

- **Forget Gate:** Decides what irrelevant past information to throw away.
- **Input Gate:** Decides what new, relevant information to add to the cell state.
- **Output Gate:** Decides what the next hidden state should be, based on the filtered cell state.

This architecture allows the keyboard app to accurately predict a word based on context established much earlier in the user's sentence.

5. Automated Customer Support (Machine Translation)

Design: Sequence-to-Sequence (Seq2Seq) Model using an Encoder-Decoder architecture.

The Role of the Encoder:

The encoder is responsible for reading and interpreting the customer's query in the source language. It processes the text word by word (or token by token) using an RNN or LSTM. With each time step, it updates its internal state. By the time it reaches the end of the sentence, it compresses the entire semantic meaning and syntactic structure of the query into its final hidden state.

Sequence Representation (Context Vector):

The final hidden state produced by the encoder is known as the Context Vector. It acts as an informational bottleneck and the bridge between the two languages. It is a fixed-size mathematical representation (a summary) of the entire input sentence. It holds the "concept" of the query independent of any specific language structure.

The Role of the Decoder:

The decoder takes this Context Vector as its initial state and begins generating the translated response in the target language. It works auto-regressively: it outputs one word at a time, and uses that generated word as the input for the next time step. It continues decoding the sequence step-by-step until it generates a special "end-of-sequence" token, completing the translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____